9.3 The Integral Test

(All series in 9.3 are assumed to have no negative terms.)

Corollary of Monotonic Sequence Theorem:

Let $\{a_n\}$ be a sequence of nonnegative real numbers. Then the series $\sum a_n$ converges if and only if the sequences $\{s_k\}$ of partial sums is bounded.

"$\Rightarrow$" If $\sum a_n$ converges to L, then the partial sums $\{s_k\}$ is bounded above by L.

"$\Leftarrow$" $\{s_k\}$ is bound & increasing.

apply MST $\Rightarrow$ $\{s_k\}$ converges

$\Rightarrow$ $\sum a_n$ converges.

Example: The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

$$\sum \frac{1}{n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \frac{1}{9} + \dots,$$

$$> \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \qquad > \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

$S_1 = 1$

$S_2 = 1 + \frac{1}{2}$

$S_4 = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + \frac{1}{2}$

$S_8 = S_4 + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) > 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

let $k_j = 2^j \Rightarrow S_{k_j} > 1 + \frac{j}{2}$. $\Rightarrow S_{k_j} \to \infty$ as $j \to \infty$

$\Rightarrow$ The partial sums $\{S_k\}$ is not bounded.

So by Corollary, $\sum \frac{1}{n}$ is a divergent series.

The Integral Test

Example: The 2-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

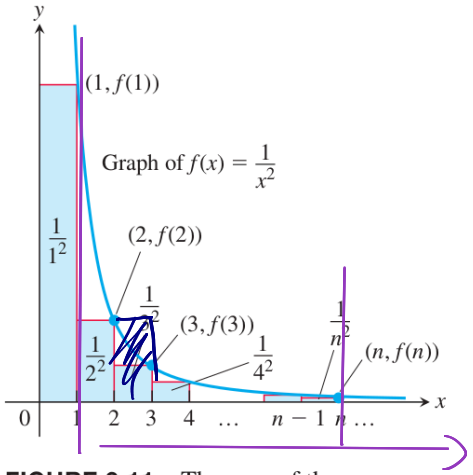


FIGURE 9.11 The sum of the areas of the rectangles under the graph of $f(x) = 1/x^2$ is less than the area under the graph (Example 2).

$a_n = \frac{1}{n^2}$

$S_n = a_1 + (a_2 + \dots + a_n)$

$\leq a_1 + \int_1^n \frac{1}{x^2}\, dx$

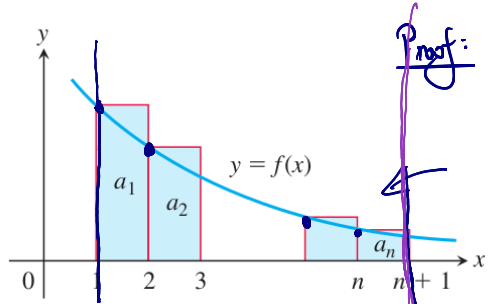
$\leq 1 + \int_1^{\infty} \frac{1}{x^2}\, dx$

$= 1 + \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b$

$= 1 + 1 - \lim_{b\to\infty} \frac{1}{b} = 2.$

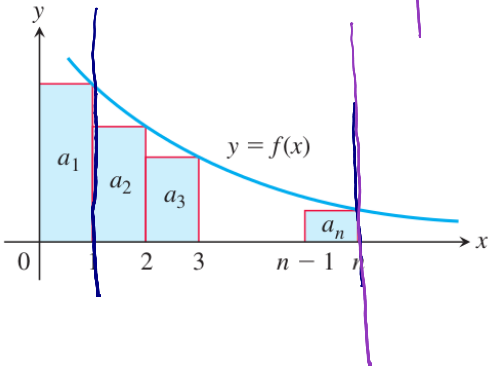
The partial sums $\{S_n\}$ is bounded above, and so by the earlier corollary, the 2-series $\sum \frac{1}{n^2}$ converges.

Theorem (The Integral Test): Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.



Proof: $(N = 1)$

$$S_n = a_1 + a_2 + \dots + a_n \geq \int_1^{n+1} f(x)\,dx$$



$$S_n = a_1 + a_2 + \dots + a_n \leq a_1 + \int_1^n f(x)\,dx.$$

$$\int_1^{n+1} f(x)\,dx \leq S_n \leq a_1 + \int_1^n f(x)\,dx$$

$$\int_1^{n+1} f(x)\,dx \le S_n \le a_1 + \int_1^{n} f(x)\,dx$$

- If $\int_1^{\infty} f(x)\,dx$ converges, then the RHS of the inequality says that $\{S_n\}$ is bounded $\Rightarrow \sum a_n$ converges.

- If $\int_1^{\infty} f(x)\,dx$ diverges, then the LHS of the inequality $\Rightarrow \{S_n\}$ is unbounded.

  $\Rightarrow \sum a_n$ diverges.

$f(x) = \frac{1}{x^p}$.

Example: Given a real constant p, the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

- $p = 1$ we have harmonic series — diverges. (or compare with $\int_1^\infty \frac{1}{x}\,dx$)

- $p > 1$: $\int_1^\infty \frac{1}{x^p}\,dx = \lim_{b\to\infty} \left[\frac{1}{1-p}\,\frac{1}{x^{p-1}}\right]_1^b$

$= \frac{1}{p-1}$ (since $\frac{1}{b^{p-1}} \to 0$ as $b \to \infty$)

- $p \leq 0$ $\frac{1}{n^p} = n^{\oplus \geq 0} \not\to 0$ nth term Test. $\Rightarrow \sum \frac{1}{n^p}$ diverges

- $0 < p < 1$ $\int_1^\infty \frac{1}{x^p}\,dx = \lim_{b\to\infty} \left[\frac{1}{1-p}\,\frac{1}{x^{p-1}}\right]_1^b$

$= \lim_{b\to\infty} \left(\frac{1}{1-p}\, b^{1-p} - \frac{1}{1-p}\right) \to \infty$.

$\therefore \sum_{n=1}^{\infty} \frac{1}{n^p}$ { converges for $p > 1$; diverges for $p \leq 1$.

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is convergent.

$f(x) = \frac{1}{x^2+1}$ cont, positive, decreasing.

compare it with $\int_1^{\infty} \frac{1}{x^2+1}\,dx$

$$= \lim_{b\to\infty} \left[\tan^{-1}(x)\right]_1^b$$

$$= \lim_{b\to\infty} \left(\tan^{-1}(b) - \frac{\pi}{4}\right)$$

$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

So by the integral test, $\sum \frac{1}{n^2+1}$ converges.

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ is divergent.

Compare with $\int_2^{\infty} \frac{1}{x \ln x}\, dx$

$(u = \ln x)$ $= \lim_{b \to \infty} \left[\ln(\ln x) \right]_2^b$

$= \lim_{b \to \infty} \left[\ln(\ln b) - \ln(\ln 2) \right]$

$\to \infty$.

By the integral test, $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges.

Error Estimation

Suppose the series $\sum_{n=1}^{\infty} a_n$ converges to S, we want to estimate the size of the **remainder** R_n between $S$and its nth partial sum s_n. That is,

$$R_n = S - s_n = a_{n+1} + a_{n+2} + a_{n+3} + \ldots$$

Let f be the function st $f(n) = a_n$,

then $a_{n+1} + a_{n+2} + a_{n+3} + \ldots \leq \int_n^{\infty} f(x)\,dx$

and $a_{n+1} + a_{n+2} + \ldots \geq \int_{n+1}^{\infty} f(x)\,dx$.

Bound for The Remainder in the Integral Test

Suppose $\{a_k\}$ is a sequence of positive numbers with $a_k = f(k)$, where f is a continuous positive decreasing function of x for all $x \geq n$, and that $\sum a_n$ converges to S. Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx.$$

$R_n = a_{n+1} + a_{n+2} + \cdots$

$S = R_n + S_n$.

To estimate S, add S_n to the inequalities

$$\Rightarrow \quad S_n + \int_{n+1}^{\infty} f(x)\,dx \leq S \leq S_n + \int_{n}^{\infty} f(x)\,dx.$$

(Recall that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$)

Example: Estimate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using the above inequalities with $n = 10$.

$a_n = \frac{1}{n^2}$, $n = 10$

$$a_{n+1} + a_{n+2} + \dots \le \int_{10}^{\infty} \frac{1}{x^2}\,dx$$

$$= \lim_{b \to \infty} \left[-\frac{1}{x} \right]_{10}^{b} = \frac{1}{10}$$

$$a_{n+1} + a_{n+2} + \dots \ge \int_{11}^{\infty} \frac{1}{x^2}\,dx = \frac{1}{11}$$

$$\Rightarrow \quad S_{10} + \frac{1}{11} \le \sum_{n=1}^{\infty} \frac{1}{n^2} \le S_{10} + \frac{1}{10}$$

$$S_{10} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{10^2} \approx 1.5498$$

$$\Rightarrow \quad \sum_{n=1}^{\infty} \frac{1}{n^2} \approx 1.6453 \qquad \left(\frac{\pi^2}{6} \approx 1.64498 \right)$$

Extra Examples:

1. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$
2. $\sum_{n=1}^{\infty} \frac{e^n}{1+e^{2n}}$
3. $\sum_{n=1}^{\infty} ne^{-n^2}$

① $\int_1^{\infty} \frac{1}{\sqrt{x}(\sqrt{x}+1)}\,dx = \int_2^{\infty} \frac{2}{y}\,dy$

let $y = \sqrt{x}+1$

$\Rightarrow dy = \frac{1}{2\sqrt{x}}\,dx$

$= \lim_{b\to\infty} \left[2\ln y \right]_2^b \to \infty$.

$\therefore$ By the Integral Test, the series diverges.

② $\int_1^{\infty} \frac{e^x}{1+e^{2x}}\,dx = \int_e^{\infty} \frac{dy}{1+y^2}$

$y = e^x$

$\Rightarrow dy = e^x\,dx$

$= \lim_{b\to\infty} \left[\tan^{-1}(y) \right]_e^b$.

$= \frac{\pi}{2} - \tan^{-1}(e)$.

Integral converges $\Rightarrow$ The series converges.

③ $\sum_{n=1}^{\infty} n e^{-n^2}$

$$\int_1^{\infty} x e^{-x^2}\, dx = \int_1^{\infty} \frac{1}{2} e^{-u}\, du$$

$u = x^2$

$\Rightarrow du = 2x\, dx$

$$= \lim_{b \to \infty} \left[-\frac{1}{2} e^{-u} \right]_1^b$$

$$= \frac{1}{2} e^{-1} < \infty$$

So by the integral test, $\sum_{n=1}^{\infty} n e^{-n^2}$ converges.

What about $\sum_{n=1}^{\infty} e^{-n^2}$? $\int_1^{\infty} e^{-x^2}\, dx$

$e^x \leq e^{x^2}$ for $x \geq 1$ $\Rightarrow$ $e^{-x^2} \leq e^{-x}$

DCT (for integral) $\Rightarrow$ $\int_1^{\infty} e^{-x^2} dx \leq \int_1^{\infty} e^{-x} dx = e^{-1}$

$\Rightarrow$ $\int_1^{\infty} e^{-x^2} dx$ converges $\Rightarrow$ $\sum_{n=1}^{\infty} e^{-1^2}$ converges